



Alberta Public Health Disease Management Guidelines

Human Immunodeficiency Virus (HIV)



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Health and Wellness Promotion Branch

Public Health and Compliance Division

Alberta Health

Human Immunodeficiency Virus (HIV) | Alberta Health, Government of Alberta

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Case Definition

Confirmed Case

Adults, Adolescents and Children ≥ 18 Months

- Detection of Human Immunodeficiency Virus (HIV) antibody with confirmation (e.g., EIA screening with confirmation by confirmatory test)

OR

- Detection of HIV nucleic acid (e.g., DNA-PCR or plasma RNA)

OR

- Detection of HIV p24 antigen with confirmation by neutralization assay (HIV-1 only)^A

OR

- Isolation of HIV in culture^B

Children < 18 Months^C

- Detection of HIV nucleic acid (e.g., DNA-PCR or plasma RNA)

OR

- Detection of HIV p24 antigen with confirmation by neutralization assay (HIV-1 only)^A

OR

- Isolation of HIV in culture

Criteria for classifying the HIV Type as HIV-2

All HIV infections in Canada are assumed to be type 1 (HIV-1) unless laboratory tests results are sufficient to classify the infection as type 2 (HIV-2) or dual HIV-1 and HIV-2 infections. Clinical or epidemiologic evidence might lead to laboratory testing for HIV-2 but is insufficient for classifying the HIV type as HIV-2.

^A Detection kits no longer available; method not currently used in AB

^B Not typically used for routine diagnosis

^C In children < 18 months of age born to HIV positive mothers, nucleic acid testing should be done within two weeks after birth and if negative repeated at one months and at four months of age. Note: Fourth generation assays (detect HIV antibodies and p24 antigens) can detect passive antibodies up to 24 months of age. (J. Robinson, personal communication March 2, 2022)

HIV Related Death^D

Any death, based upon clinical knowledge of the patient, which can be attributed directly or considered as a contributing factor to confirmed HIV infection. May include, but not limited to Acquired Immune Deficiency Syndrome (AIDS) defining illnesses^E and non-AIDS morbidity.

^D The reporting of an AIDS diagnosis is no longer required in Alberta

^E A list of indicator diseases can be found at <https://www.canada.ca/en/public-health/services/diseases/hiv-aids/health-professionals/national-case-definition-aids.html>

Reporting Requirements

Physicians, Health Practitioner and Others

Physicians, health practitioners and others listed in Section 22 of the [Public Health Act](#) shall notify the Sexually Transmitted Infection (STI) Medical Director via Sexually Transmitted Infection Centralized Services (STICS), of all confirmed cases by forwarding a completed [HIV Case report form](#) or its data elements by phone, fax, or electronic transfer within 48 hours (two days).

Laboratories

- All laboratories shall report all positive laboratory results by mail, fax or electronic transfer within 48 hours (two business days) to the STI Medical director via STICS and the Chief Medical Officer of Health (CMOH) (or designate).

Alberta Health Services

Contact Information: **Toll free: 1-855-945-6700 option 4**
 Fax: 780-670-3624

The STI Medical Director/STICS shall forward the preliminary [Alberta HIV Case Report](#) form of all confirmed cases to the CMOH (or designate) within four weeks of notification and the final amendments within ten weeks of notification. The STI Medical Director /STICS are responsible for ensuring investigation and follow-up of all reported confirmed cases.

- For out-of-province and out-of-country reports, the following information should be forwarded to the CMOH (or designate) by the mutually agreed upon reporting system within 48 hours (two days) including:
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - attending physician (locally and out-of-province), and
 - positive laboratory report (faxed).
- For out-of-province and out-of-country contacts, the following information should be forwarded to the CMOH (or designate) by the mutually agreed upon reporting system within 48 hours (two days):
 - name,
 - date of birth,
 - personal health number, and
 - contact information (i.e., address and phone number).

Canadian Blood Services (CBS)

- All newly diagnosed HIV positive persons **must** be reported by the STI Medical Director (or MOH designate) to CBS as soon as possible if: the person has a history of donating blood/blood products (M. Bigham, personal communication, March 16, 2022). A copy of the positive test result must accompany the Transmissible Disease Notification (TDN) form, and all information should be sent for Lookback/Traceback to the TDN Department by:
 - Fax 1-844-836-6843 **OR**
 - Scan and email to: TDnotifications@blood.ca
- To speak to a TDN Specialist regarding any questions or concerns please call (506) 648-5076 or 1-888-992-5663 ext. 5076.

Citizenship and Immigration Canada (CIC)

- Out-of-country immigrants, arriving in Alberta and testing positive for HIV, may be notified to Alberta Health Services (AHS) by CIC, dependent upon class and consent of immigrant.
- Alberta Health will forward a list of any HIV positive immigrants, arriving in Alberta, received from CIC to AHS.

Additional Reporting Requirements

In all cases, where a person under 18 is suspected or confirmed to have a new diagnosis of HIV, an assessment should be carried out by the clinician to determine if additional reporting is required.

To Alberta Child and Family Services

- The clinician should determine whether there are reasonable and probable grounds to believe that they are in contact with “a child in need of intervention” (as per Section 1(2) of the [Child, Youth and Family Enhancement \(CYFE\) Act](#)) and shall report to a director pursuant to Section 4 of the *CYFE Act*.⁽¹⁾
- Reporting is done by contacting the local Child and Family Services office or calling the **CHILD ABUSE HOTLINE: 1-800-387-5437 (KIDS)**. For local office contact information, see [the Children's Services Ministry's website](#).

To Law Enforcement Agency

- Consent is a key factor in determining whether any form of sexual activity is a criminal offence. Children under 12 do not have the legal capacity to consent to any form of sexual activity. The law recognizes that the age of consent for sexual activity is 16. However, the law identifies the exception for minors between 12 and 16 years as having the ability to consent, in “close in age” or “peer group” situations. ⁽²⁾
- Reporting is done by contacting your local City Police Detachment or [RCMP Detachment](#).
- For additional information:
 - Alberta [Child, Youth and Family Enhancement Act](#)
 - Age of Consent to Sexual Activity at www.justice.gc.ca/eng/rp-pr/other-autre/clp/faq.html
 - Criminal Code of Canada at www.laws-lois.justice.gc.ca/eng/acts/C-46/ ⁽³⁾

Epidemiology

Etiology

HIV is a retrovirus of which two types have been identified: type 1 (HIV-1) and type 2 (HIV-2). These viruses are serologically and geographically distinct and have different efficiencies of transmission, disease progression and mortality.^(4,5) Globally and in Alberta HIV-1 is the predominant virus, with HIV-2 being extremely uncommon.⁽⁶⁾

Clinical Presentation

Adults

Within seven to 10 days of infection with HIV, approximately 30–70% of persons usually develop a non-specific influenza-like illness (i.e., fever, malaise, pharyngitis, anorexia, weight loss, lymphadenopathy, fatigue) that may last an average of two weeks.⁽⁷⁾ Untreated, the average time to the development of an AIDS-defining illness is eight to 15 years.⁽⁷⁾ With current highly active antiretroviral therapy (HAART) regimes and appropriate management, the life expectancy of a person living with HIV (LWHIV) and those un-infected can be similar in North America.⁽⁸⁾ Increasingly, HIV infection is considered a chronic disease for persons who have access and adhere to therapy.⁽⁹⁾

Clinical illness may include opportunistic infections (e.g., *Pneumocystis jirovecii* [formerly *Pneumocystis carinii*] pneumonia, disseminated *Mycobacterium avium* complex (MAC) disease, primary neurologic disease (e.g., AIDS dementia) and malignancy (e.g., lymphoma, Kaposi sarcoma).^(10,11) Effective early treatment reduces the mortality related to HIV and progression to an AIDS-defining illness.^(9,10,12,13)

Children

The median time to disease progression of untreated perinatally-infected children is 3 years. Untreated children commonly present with Pneumocystis pneumonia but other features can include poor growth, diarrhea, enlarged lymph nodes, developmental delay and recurring bacterial infections⁽¹⁴⁾. Up to 15% of perinatally-infected children who are untreated will not survive beyond 18-36 months. With treatment, disease progression can be delayed indefinitely.^(14–16)

Diagnosis

Ages >18 months

In Alberta, 4th generation assays (enzyme-linked immunosorbent assay (ELISA) to detect both HIV antibodies and p24 antigen) are used by all clinical diagnostic laboratories for screening for HIV-1 and HIV-2 although the majority of testing is performed at the Public Health Laboratory (ProvLab). Private laboratories perform third-party HIV tests, and all positive tests are forwarded to ProvLab for confirmation. Confirmation of the positive screen is through an immunochromatographic confirmatory assay that detects and distinguishes between antibodies to HIV-1 and HIV-2. Samples that are positive in the confirmatory assay are reported as positive.

The p24 component (HIV-1 core antigen) of the screening assay, can detect early acute HIV-1 infection scenarios, i.e. when antibody is absent or low and HIV-1 viral load is high. In these circumstances, the confirmatory serology assay will yield either a negative or an indeterminate result. A follow up blood test for HIV viral load is recommended and requested by ProvLab to assess if this is an acute infection or a false positive results. Repeat serology testing is also recommended after 2 weeks to assess the HIV status of the individual. (K. Fonseca, personal communication, December 2021)

With the current 4th generation assays, the “window period” for HIV detection has been reduced to about 2 weeks after infection, although retesting at 3 months post exposure is still recommended in those cases where initial test is negative and seroconversion may be slower (such as when the subject is immunocompromised), or is co-infected with Hepatitis C. ^{(17,18),(J. Robinson personal communication March 2, 2022)}

Oral pre-exposure prophylaxis (PrEP) has also been shown to increase the time period to serologic seroconversion for individuals taking this medication.⁽¹⁹⁾ Hence, consultation with an infectious disease specialist or the laboratory is recommended in these cases.⁽²⁰⁾

Ages <18 months

For newborns from HIV-infected mothers, nucleic acid testing (NAT) by DNA PCR (proviral DNA) or plasma RNA are required to make a diagnosis of vertical transmission instead of serologic testing as the maternal HIV IgG antibodies cross the placenta and give a positive serologic result likely reflective of mother's status. The timing of nucleic acid testing depends upon mother's HIV status or the risk of transmission but is most often indicated at birth, then at 1- months and 4 months.⁽²¹⁾ All positive NAT tests should be repeated with a second specimen for confirmation. For children with negative NAT results, antibody testing should be performed at 24 months to ensure they have lost maternally derived antibodies. (K. Fonseca, personal communication, Dec 2021, J. Robinson, personal communication March 2022)

Role of rapid and other testing technologies

Advances in technologies have provided opportunities for the role of rapid and point of care (POC) testing. The INSTI® HIV Self Test is available in Alberta health care settings where a rapid test is indicated for more immediate management of patients.⁽²²⁾ Further, at home self-tests and community based POC testing affords a greater opportunity for screening in remote and/or hard to reach populations, reduce barriers to care and increase linkages to care, among other benefits.⁽²³⁾ Positive results from any of these testing modalities do not meet case definition and legislative reporting requirements. Regardless of testing format, all reactive and indeterminate results must be confirmed by traditional serological lab testing to meet legislated reporting requirements.⁽²⁴⁾

Dried Blood Spot (DBS) testing is an alternate specimen collection method for HIV. The specimen undergoes the same 4th generation testing process as a serological sample. However, reactive DBS results are subsequently confirmed by standard serological test processes before they are reported as the former process is not approved by Health Canada.

Treatment

- This is an increasingly complex area with rapid changes in optimal therapy as new research becomes available.
- Recommendations should be made in collaboration with a physician experienced in HIV/AIDS care and treatment.
- Encourage and support people who are HIV positive to take the medications prescribed for them. A directly observed treatment (DOT) program may be considered for populations who have unstable social situations and where such programs are available.
- Positive pregnant women should:
 - be advised of overwhelming benefit of antiretroviral drugs in preventing perinatal transmission, and
 - receive antiretroviral therapy prenatally from the day of diagnosis (ideally prior to pregnancy) and during labor and delivery.

Reservoir

Primarily humans, although closely related strains continue to circulate in non-human primates.^(11,15)

Transmission

Established routes of transmission are:

- sexual contact,^F
- percutaneous contaminated blood exposure,
- mucous membrane exposure to contaminated blood,

^F Sexual contacts includes vaginal, anal or orogenital direct contact.

- mother to child transmission (MTCT)⁶, and
- transfusion with contaminated blood products⁽¹⁵⁾

The HIV virus is most commonly found in and transmitted through blood, body fluids containing blood and other body fluids (i.e., semen) with a high viral titre.⁽²⁵⁾ It has been isolated from urine, saliva, tears, and bronchial secretions, however, transmission from these fluids has not been documented and is considered largely improbable.^(11,26,27) Concurrent sexually transmitted infection (STI), especially ulcerative STI, greatly facilitates the transmission and acquisition of HIV.⁽²⁵⁾ Infection may be transmitted vertically from mother to child during pregnancy, delivery or through breastfeeding.^(11,15) However, MTCT is significantly decreased with the use of HAART.⁽²⁸⁾ Women who achieve viral load suppression prior to the onset of pregnancy will transmit <1% of the time.^(28–30)

Person with HIV infection who have achieved viral suppression (< 200 copies/ml) by using HAART cannot sexually transmit the virus to others.^(31–33) This is referred to as ‘undetectable equals untransmittable’ (U=U), where effective HIV treatment is considered preventive.⁽³⁴⁾

Incubation Period

The incubation period is variable. The time frame from infection to detectable antibodies can range from two to three weeks to four months, although 95% of persons are detectable within 35 days.^(17,35,36) Time to detection of HIV after infection is also variable depending upon testing method.⁽³⁷⁾ Time from HIV infection to development of AIDS has an observed range of 1-15 years without treatment.⁽¹¹⁾ Untreated, risk of progression to AIDS is about 1 to 2% per year in the first 2 to 3 years of infection and about 5 to 6% per year thereafter. Eventually, AIDS almost invariably develops in untreated patients.⁽³⁰⁾

Period of Communicability

Epidemiological evidence suggests that transmissibility begins early after the onset of HIV infection.^(15,31) Infectiousness is highest during the initial infection and rises with increasing immune deficiency. However, subjects with undetectable viral loads are not thought to transmit through sexual activity.^(31,33)

Host Susceptibility

Persons are universally susceptible; however, existing chlamydia and/or gonorrhea infection can increase risk of acquisition. The presence of other STIs, especially ulcerative STIs such as syphilis, also increases the likelihood of transmission.^(10,12,13,25)

Incidence

General

AIDS was first reported as a novel syndrome in 1981; two years later, the viral etiologic agent was isolated.⁽³⁸⁾ AIDS has been documented in virtually all countries of the world, among all races, ages, and social classes. The Joint United Nations Programme on HIV/AIDS (UNAIDS) estimated that in 2020, globally, 37.7 million people were living with HIV, 1.5 million became newly infected, and 680 000 people died of AIDS.⁽³⁹⁾

AIDS and HIV became reportable in Alberta in 1983 and 1998 respectively.

Refer to the [Interactive Health Data Application](#) for more information.

⁶ MTCT includes: in utero, prenatally, postnatally and breastfeeding.

Public Health Management

Key Investigation

- AHS employees: refer to the internal document *HIV Practice Support*
- Determine the reason for the test (from the case or health care provider).
- Assess potential risk factors for infection:

Non-Sexual

- injection drug use (including steroids),
- sharing of needles or other paraphernalia
- receipt of blood/tissue/organ between 1978 and 1985,
- receipt of blood/tissue/organ at any time in a developing country,
- skin piercing procedures (e.g., tattooing, body piercing, acupuncture),
- workplace or occupational exposure,
- recent invasive medical or dental procedures.

Sexual

- Assess sexual relationships and high-risk sexual behaviors including:
 - all persons with new, anonymous or multiple sex partners,
 - sexual assault,
 - sexual partner of a person with confirmed HIV,
 - persons and patrons of those who exchange goods/services for sex,
 - sex with partners from a HIV-endemic country or with partners with any of the above non-sexual and sexual risk factors.
- Ascertain status of co-infection with other sexually transmitted infections (STIs) and blood borne infections (BBIs) and last negative HIV test
- If female, determine pregnancy status.
- Determine donation history of blood, tissue, organs, bodily fluids i.e., milk, semen.
- Identify household and other intimate contacts for potential blood exposure from the case. Partners should be traced based on estimated duration of infection in index case.

Identify Management of a Case

- Dedicated partner notification nurses (PNNs) with STICS will contact the physician within two working days of receipt of positive test result to determine:
 - if client notified of test result,
 - public health follow-up including client education,
 - follow-up of contacts,
 - obtaining additional epidemiological information, and
 - risks associated with other STIs, hepatitis B and C to direct further testing.
- Educate the case about the modes of transmission and reducing the risk of transmission to others, including informing the case about the duty to disclose status to sexual and/or non-sexual partners.
- Screen for TB based on the [Canadian Tuberculosis Standards](#)
- Encourage engagement with regular follow-up with a physician experienced in HIV/AIDS care and treatment.
- Offer immunization to this at-risk group as recommended in the current [Alberta Immunization Policy](#)

Management of Contacts

Reference 40 applies to this section.

- It is a public health responsibility to ensure that partner notification and follow-up take place. All HIV-positive individuals are assumed infectious and capable of transmitting the virus through exchange of blood and body fluids. They must, therefore, be interviewed to identify and disclose names of their contacts. Any identifying information of the source case is held completely confidential by PNNs.
- Give priority for follow-up to pregnant female contacts.
- Tracing of partners should be based on the estimated duration of infection. If the date of seroconversion is known, all partners in the six months prior to the positive testing should be identified. If the seroconversion date is unknown, all partners, as far back as practical, should be identified. (minimum 1 year or what is reasonable)
- Personnel conducting contact tracing should always provide partners with information that includes:
 - modes of transmission,
 - disease process,
 - how to modify risk behaviors, and
 - telephone numbers and addresses of support agencies and testing clinics.
- All partners should be asked about their previous HIV testing and encouraged to be tested for this HIV exposure and given specific details on where to be tested, and how it will be reported if positive.
- Information on HIV PEP and PrEP should be provided as applicable with information on how to access
- Other contacts: Individuals exposed to blood and other body fluids capable of HIV infection should be:
 - notified of potential HIV exposure,
 - assessed for PEP, and
 - educated about eligibility for PrEP.

Care of neonate

- Refer infants born to HIV-positive women as soon as possible after delivery to a specialist in pediatric infectious diseases for assessment, establishing serological testing frequency and appropriate HIV prophylaxis and treatment.
- Breastfeeding is not recommended for HIV-positive mothers.⁽⁴¹⁾

Preventive Measures

- Prevention and public health programs should be offered to reduce HIV transmission in people who inject drugs (e.g., needle exchange programs and harm reduction strategies).
- HIV post-exposure prophylaxis (PEP) should be considered for occupational, non-occupational exposures and sexual assaults as outlined in the [Alberta Guidelines for Post-Exposure Management and Prophylaxis: HIV, Hepatitis B, Hepatitis C and Sexually Transmitted Infections](#). When eligibility criteria are met, the antivirals are available free of charge from Alberta Health. (PEP medication costs for Occupational exposures are the responsibility of the employer) HIV pre-exposure prophylaxis (PrEP) should be made available to eligible HIV negative persons to reduce or eliminate risk of new HIV infections according to the [Alberta PrEP Guidelines](#).
- HIV testing should be made universally available in facilities where populations are more vulnerable to HIV infection.
- Health care practitioners should include HIV testing with all routine STI screening, at least on an annual basis for individuals involved in high risk practices.⁽¹⁰⁾
- HIV testing is recommended for all pregnant women, according to the [Alberta Prenatal Screening Guidelines](#).
- Screen all donations of blood, blood products, tissues, organs, semen and breast milk for HIV.
- Provide public education about the safe handling of blood, body fluids, and sharps disposal.
- Focus on methods to reduce high risk sexual behaviors that may lead to HIV or STIs (e.g., safer sex education).
- School curricula should center on basic and accurate information about STIs, safer sex, HIV, and unplanned pregnancies.
- Family physicians should be targeted for education to increase testing and to increase awareness about the changing epidemiology of HIV/AIDS.
- Skin piercing procedures – Anyone considering tattooing, body piercing, or acupuncture should be counselled to ensure that these practices are carried out with sterile equipment, preferably single-use equipment.

Note: PLWHIV who are virally suppressed via antiretroviral therapy are considered non-infectious and not capable of sexual transmission and substantially reduce the risk of perinatal and breastfeeding transmission.^(31,33,34)

Individuals Unwilling or Unable to Follow Transmission Prevention Practices

- Educate the individual about the modes of transmission and reducing the risk of transmission to others, and their public health and legal responsibility (duty to disclose to sexual and/or non-sexual partners).
- People who are unwilling or unable to take appropriate precautions to prevent the spread of HIV are to be reported to the MOH for the zone. This requires assessment by the HIV designate nurse.
 - These individuals should be handled as per regional protocol.
 - For further information refer to the [Public Health Act, Section 39\(1\)](#).⁽⁴²⁾

Health Care Workers

- Under the Health Professions Act (HPA), all health professionals are governed by regulatory colleges. They are required to govern in a way that protects and serves the public interest. Accompanying standards of practices and codes of ethics guide the HCW responsibility for ensuring their competency to practice (HPA).⁽⁴³⁾ HIV positive HCWs who are involved in exposure prone procedures (EPP) (shall contact the Zone MOH or designate to discuss the potential risks of transmission to clients. Upon assessment by the Zone MOH, a worker may or may not be referred to the Alberta Expert Review Panel for Blood Borne Viral Infections in Health Care Workers (administered by the Alberta College of Physicians and Surgeons) for further assessment services, if indicated.
- The Panel's mandate is to review circumstances of HCWs who are found to have a blood borne viral infectious disease. The panel may receive referrals from MOHs regarding HCWs who perform exposure-prone procedures when there is uncertainty as to whether continued or modified professional practice is indicated.
- In general, the recommendations for management of a HIV infected HCW are outlined by the Public Health Agency of Canada's document, [Guidelines on the Prevention of Transmission of Bloodborne viruses from Infected Healthcare Workers in Healthcare Settings](#).

Appendix 1: Revision History

Revision Date	Document Section	Description of Revision
November 2021	General	- Updated Template - Etiology, Clinical Presentation, Diagnosis and Treatment sections moved to Epidemiology - Key Investigation section moved to Public Health Management (formerly called Control) - Updated web links - Added web links where applicable
	Reporting Requirements	Updated to reflect current processes
	Incidence	Added weblinks to surveillance sites.
July 2022	Case Definition	Additional content regarding HIV-2 and HIV related death
	Reporting Requirements	Updated to reflect current processes
	Epidemiology, Public Health Management	All sections modernized with current research and program development and/or weblinks

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